Supermarkets can stock around 90 000 different products. How do you as the customer, find what you want?



Imagine if shops grouped their products into only two categories such as 'fresh' and 'preserved' or 'vegetable' and 'meat'. It could take a long time to find a tin of baked beans!

Of course, supermarkets are much more sophisticated than that, after-all they want happy shoppers at the end of the day.

Just like supermarket bosses, scientists like to order things into groups. This is called **classification**. The natural world contains maybe 8.7 million species of plants, animals and micro-organisms. That's a lot of organisms to sift through if you want to find out what the creature you saw at the sea-side is.

Use the information provided to reveal how developments in science and technology have enabled scientists over the years to produce a more refined method of classifying living organisms.

Task

1. Use Table 1 and some graph paper to help you construct an accurate timeline for the history of classification.

    - What do you notice about your timeline?
    - Why do you think this is?

2. Add the developments outlined in Table 2 to your timeline.
   Choose a development and describe the influence it had on classification.

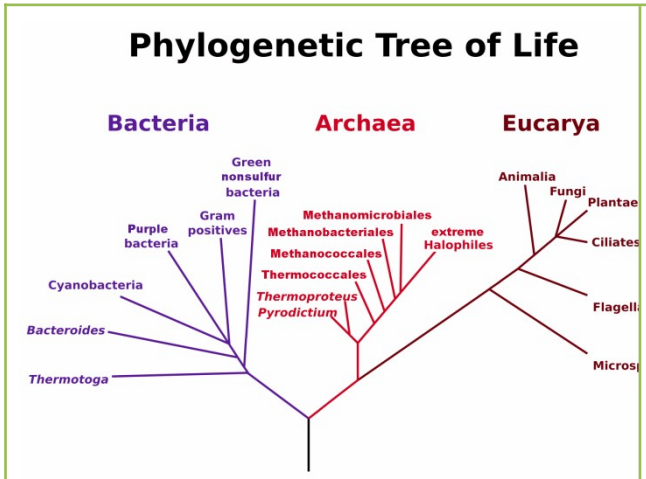
3. Cut out the images provided or find your own and add them to your timeline to illustrate the methods of classification.

Table 1: Important dates in the development of a method of classification for living things

Scientist	Date	Method of classification
Aristotle	384-322 BC	All living things divided into **plants** and **animals**. Animals were further divided into **swimming**, **flying** and **walking**.
Andrea Cesalpino	1519-1603	Classified plants according to their **fruits** and **seeds**.
Carl Linnaeus	1707-1778	Grouped organisms into **two kingdoms**: **plants** and **animals**.
Ernst Haeckel	1870's	Produced the first example of an **evolutionary tree (called 'The tree of mammals')**.
Edouard Chatton	1937	In 1937 Edouard Chatton divided organisms into **prokaryotes** and **eukaryotes**. His work was forgotten until the 1960s.
Many scientists	1950s	Scientists agreed to have **five kingdoms**: **plants**, **animals**, **fungi**, **protists** and **bacteria**.
Carl Woese	1977	In 1977 Carl Woese proposed a '3-domain system' of **Archaea** (primitive bacteria living in extreme environments), **Bacteria** (referred to as true bacteria) and **Eukaryotes** (all other organisms including protists, fungi, plants and animals).

Table 2: Important developments which have contributed towards changes in the method of classification

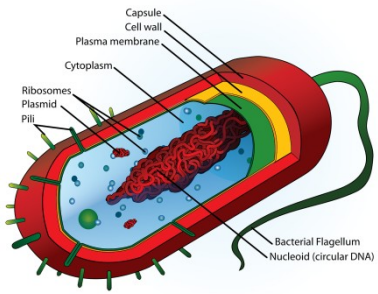
Date	Development
1400s	Plants were studied for their medicinal properties which were described in books known as **Herbals**. There was little interest in classifying animals.
1650s	The first **light microscopes** to magnify objects 266 times, were produced.
1859	Darwin's 'The Origin of Species' was first published. This explained his scientific **theory** that organisms change and **evolve** over time by a process of **natural selection**.
1930s	The first **electron microscopes** were developed**.** These enabled scientists to study the structures inside the cells of plants, animals and bacteria. It revealed differences between the cells of bacteria and other organisms.
1970s	The process of **electrophoresis** enabled scientists to sequence the **DNA** of an organism. The evolution of the **genetic code** could now be investigated.



Tree of life: three-domain system.



Dolphin



Bacterial cell



Animals



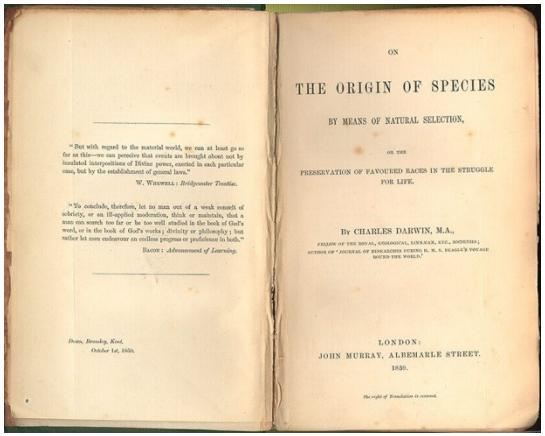
Herbal



Electron microscope



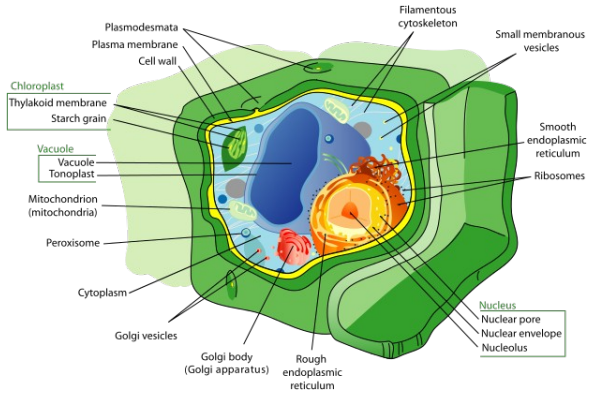
Fungi



Origin of species



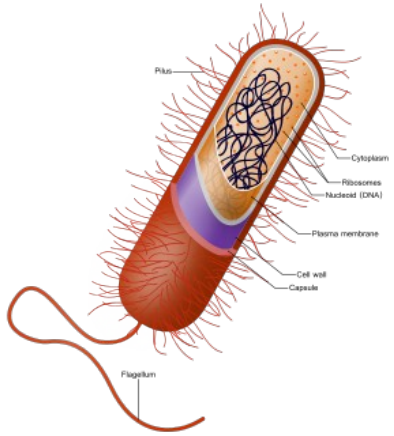
Trees



Plant cell (eukaryotic)



Fruits and seeds



Prokaryote

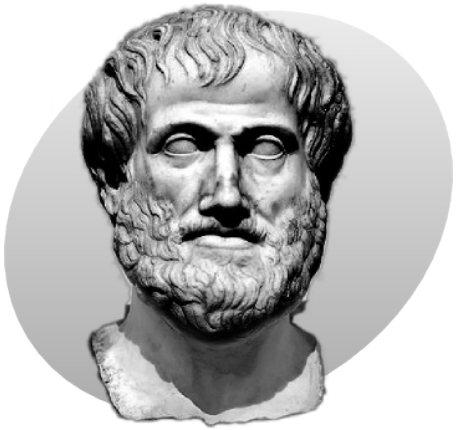
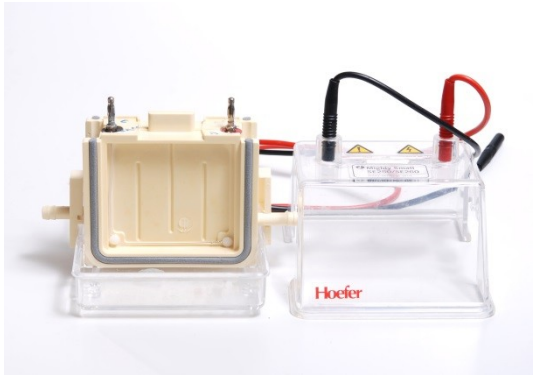
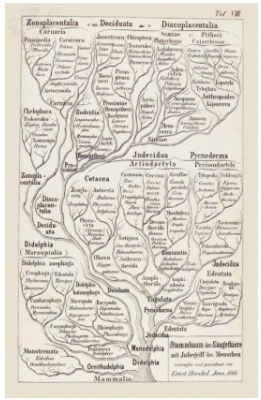
		
Aristotle	**Electrophoresis**	**Trees**
		
Bat	**Tree of mammals**	**Three major figures in light microscopy**

Image credits

- © Con-struct (Own work) [CC BY-SA 3.0 (http://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons, commons.wikimedia.org/wiki/File%3AP_Aristotle_grey.png
- © Maulucioni (Own work) [CC BY-SA 3.0 (http://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons, commons.wikimedia.org/wiki/File%3APhylogeneticTree%2C_Woese_1990.PNG
- © Lilly_M [GFDL (http://www.gnu.org/copyleft/fdl.html) or CC BY-SA 3.0 (http://creativecommons.org/licenses/by-sa/3.0)], via Wikimedia Commons, commons.wikimedia.org/wiki/File%3APolyacrylamid_gel_electrophoresis_apparatus-02.jpg
- Herbal. Herbal. Italy, N. (Lombardy). c. 1440. Italy, N. (Lombardy). From The National and Domestic History of England / Credit: British Library / Universal Images Group / Copyright © The British Library / For Education Use Only. This and millions of other educational images are available through Britannica Image Quest. For a free trial, please visit www.britannica.co.uk/trial
- Tree of mammals, historical artwork, 1866 From The National and Domestic History of England / Credit: MEHAU KULYK / SCIENCE PHOTO LIBRARY / Universal Images Group / Copyright © Science Photo Library / For Education Use Only. This and millions of other educational images are available through Britannica Image Quest. For a free trial, please visit www.britannica.co.uk/trial
- Three major figures in light microscopy From The National and Domestic History of England / Credit: JOHN READER / SCIENCE PHOTO LIBRARY / Universal Images Group / Copyright © Science Photo Library / For Education Use Only. This and millions of other educational images are available through Britannica Image Quest. For a free trial, please visit www.britannica.co.uk/trial